AI Cybersecurity Framework – One-Page Overview Map

AI-Specific Threats

- Adversarial Attacks: Data poisoning, evasion
- Model Theft/Extraction: IP loss
- Data Leakage: Inference attacks
- Bias Exploitation: Unfair outcomes

Aligned Standards

- NIST AI RMF: Risk Management Framework
- ISO/IEC 42001: AIMS
- OWASP AI Security Top 10
- GDPR/CCPA: Privacy regulations

Govern

- Objectives: Establish policies, assign accountability, risk tolerance.
- Core Controls: AI governance board, ethical guidelines, compliance audit.
- Outputs: Approved policies, governance structure, risk appetite statement.

Identify

- Objectives: Map assets, understand data, assess AI risks.
- Core Controls: Asset inventory, data lineage, threat modeling.
- Outputs: Risk register, asset inventory map, threat profiles.

Protect

- Objectives: Implement safeguards, secure data & models.
- Core Controls: Encryption, access control, secure development lifecycle.
- Outputs: Secure architecture, encrypted data, trained personnel.

Detect

- Objectives: Continuous monitoring, identify anomalies.
- Core Controls: Model monitoring, log analysis, threat hunting.
- Outputs: Alerts, anomaly reports, threat intelligence.

Respond

- Objectives: Contain incidents, minimize impact.
- Core Controls: Incident response plan, playbooks, forensics.
- Outputs: Incident report, mitigation actions, lessons learned.

Recover

- Objectives: Restore services, improve resilience.
- Core Controls: Backup & recovery, business continuity, post-incident analysis.
- Outputs: Restored systems, improved recovery plans, resilience report.

AI SYSTEMS

Cross-Cutting Principles

Ethics & Fairness: Bias mitigation, transparency.

Explainability & Interpretability: Model understanding.

Human-in-the-Loop: Oversight, accountability.

Privacy & Security: Data prota protection by design.